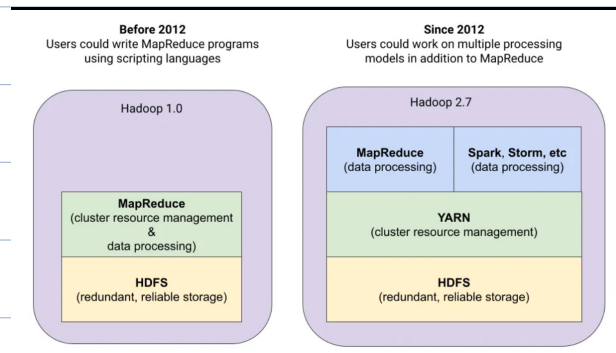
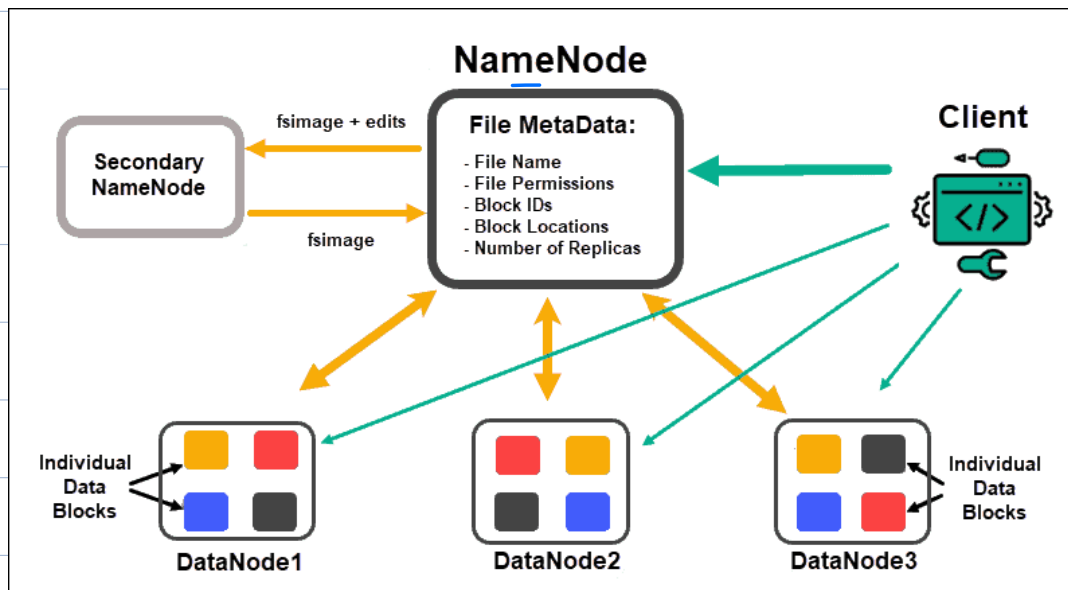


Hdfs & Yarn Architecture

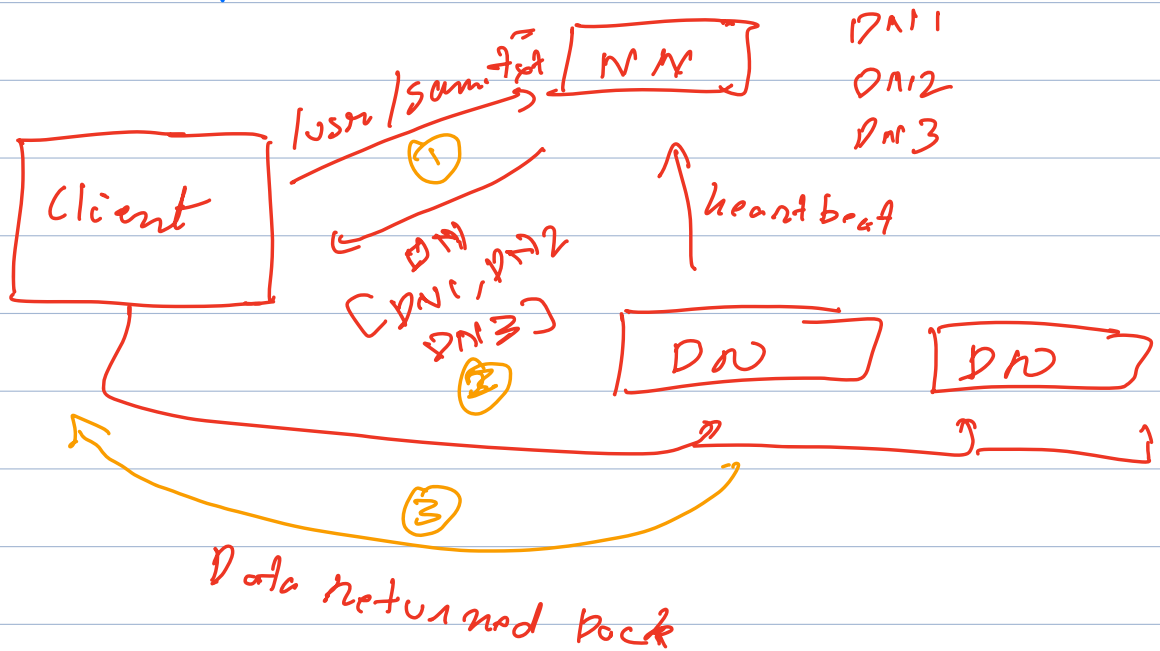


Hdfs Architecture



Hdfs cluster
1 node

Replication factor



Hdfs Operations

① list of files $\Rightarrow$ NN only

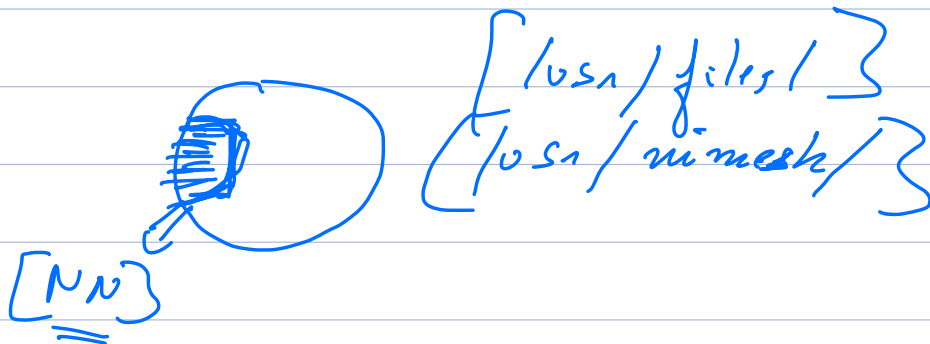
hdfs dfs -ls /user/

② Create a directory $\Rightarrow$ NN only

③ Writing a data $\Rightarrow$ NN & DN

④ Read the data $\Rightarrow$ DN & NN

⑤ Cut / paste $\Rightarrow$ Move my data $\Rightarrow$ NN

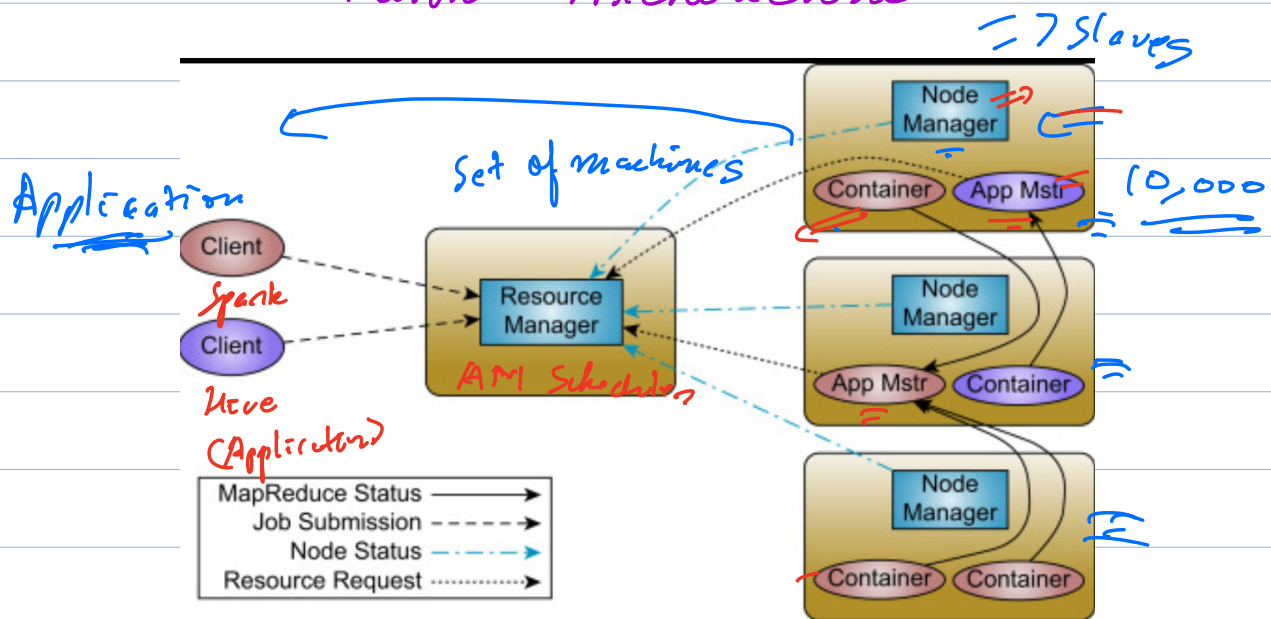


NN
B1 D1 /usr/files $\Rightarrow$ B1 D1 /usr/nimash

⑥ Copy No data $\Rightarrow$ DN & NN

B1 D1 /usr/files
~~B2~~ ~~D2/D1~~ /usr/nimash

Yarn Architecture



Resource Manager $\Rightarrow$ Master component

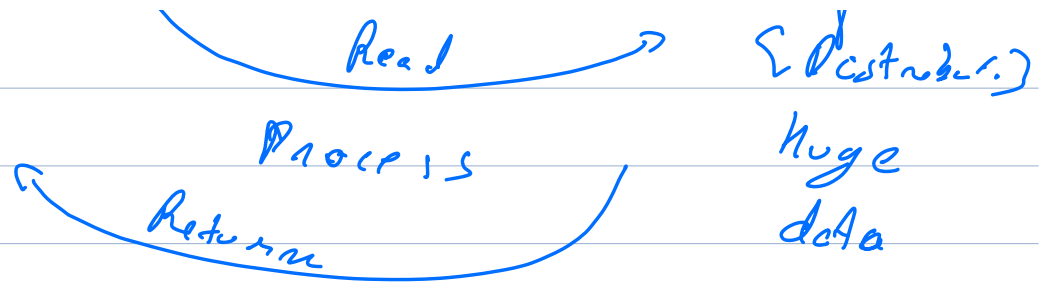
- $\hookrightarrow$ Scheduler $\Rightarrow$ ①
- $\hookrightarrow$ Application Manager ②

Resource Manager

Hive / Spark $\xrightarrow{\text{SQL}}$ (Application)

$\hookrightarrow$ select from table $\rightarrow$ 1 Application

$\hookrightarrow$ tasks



Scheduler $\Rightarrow$ FIFO, capacity scheduling

1000 app

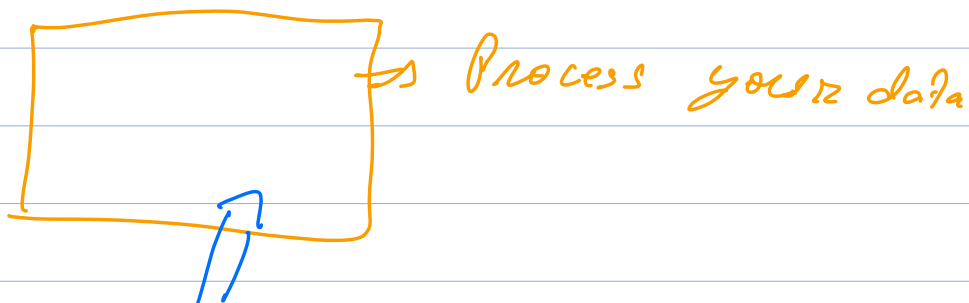
10,000

10 machines

$\rightarrow$

Application Manager $\Rightarrow$ 1,00,000
↳ talks to applications
↳ Scheduler (wait)

Physical Nodes



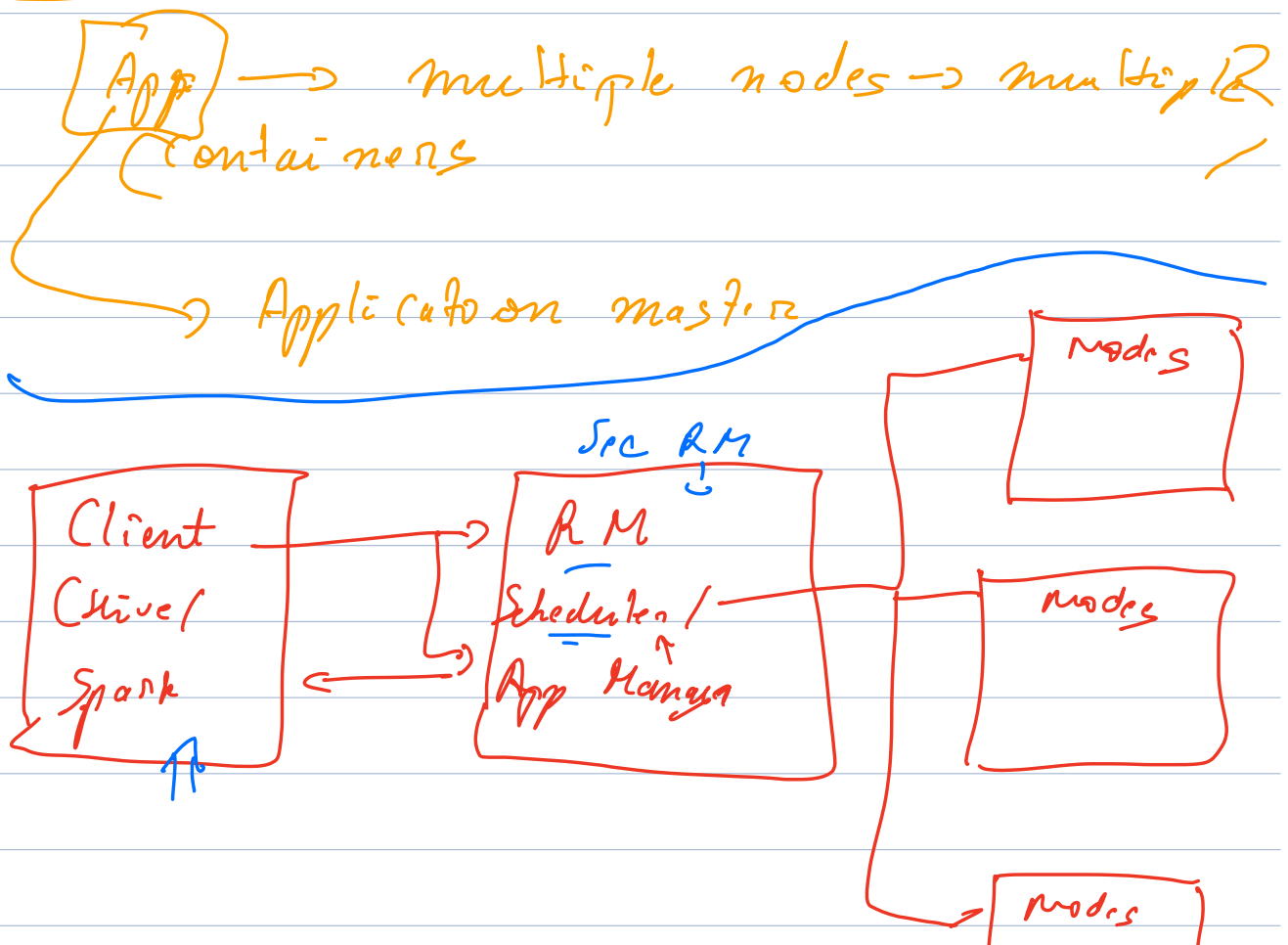
[Node Manager] ^{or}

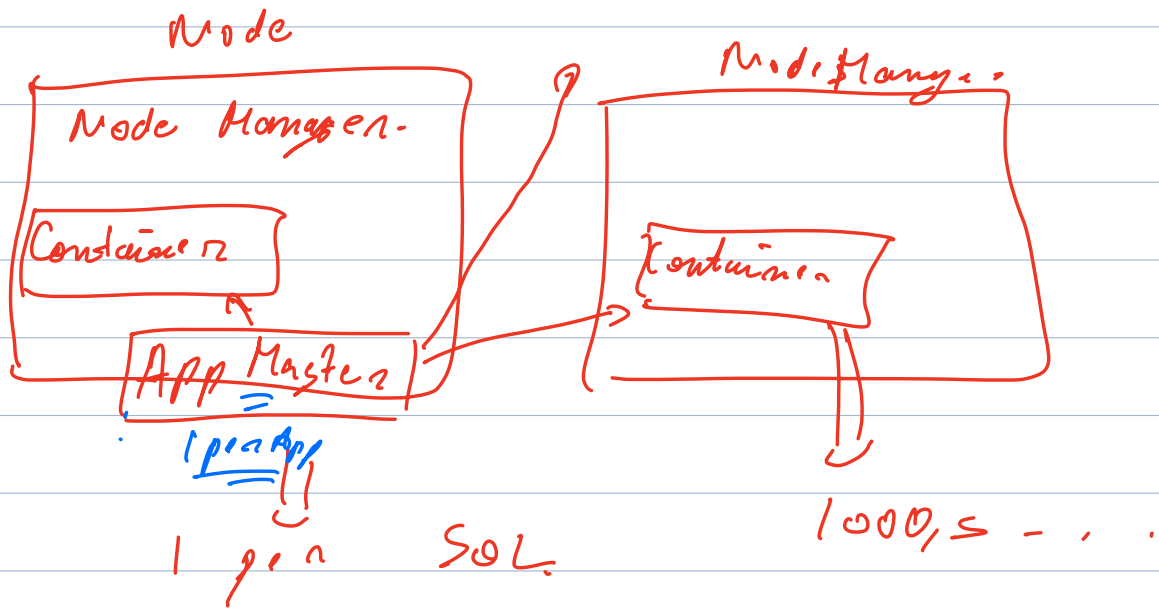
⇒ Heart beat

⇒ Machine is working fine [status of machine]

[Application Master]

Application level monitoring → Yarn





Tb, Petabytes

SPOF → Single Point of Failure.

